Statistical Learning

Week 12

Lecture: Monday, October 12, 2026

1 Multi-class Classification via Dummy Encoding

Consider covariates $X_1, X_2, \ldots, X_d$ with a categorical class label taking values in $\{A, B, C\}$:

X_1	X_2	X_3	$\cdots$	X_d	Class label
					A
					B
					C
					B
					C
					A
					C
					B

Two encoding options are possible.

One option (2 dummy columns)

Class	Y_1	Y_2
A	1	0
B	0	1
C	0	0
B	0	1
C	0	0
A	1	0
C	0	0
B	0	1

Second option (full one-hot encoding)

Class	Y_1	Y_2	Y_3
A	1	0	0
B	0	1	0
C	0	0	1
B	0	1	0
C	0	0	1
A	1	0	0
C	0	0	1
B	0	1	0

With the full one-hot encoding:

$$\widehat{P}(A \mid x_1 \cdots x_d) = \widehat{E}(Y_1 \mid x_1 \cdots x_d), \qquad \widehat{P}(B \mid x_1 \cdots x_d) = \widehat{E}(Y_2 \mid x_1 \cdots x_d), \qquad \widehat{P}(C \mid x_1 \cdots x_d) = \widehat{E}(Y_3 \mid x_1 \cdots x_d$$

A classification rule can be formed based on these estimates (using the 2-column encoding):

$$\widehat{E}(Y_1 \mid x_1 \cdots x_d) = \widehat{P}(A \mid x_1 \cdots x_d), \qquad \widehat{E}(Y_2 \mid x_1 \cdots x_d) = \widehat{P}(B \mid x_1 \cdots x_d)$$

$$\widehat{P}(C \mid x_1 \cdots x_d) = 1 - \widehat{P}(A \mid x_1 \cdots x_d) - \widehat{P}(B \mid x_1 \cdots x_d)$$

How also a classifier can be formed based on these estimates.

Still there two methods lead to the same decision rule?

$$\sum_{j=1}^{3} \widehat{E}(Y_j \mid x_1 \cdots x_d) = 1 \quad ?$$

1.1 Kernel (Nadaraya–Watson) Regression

$$\frac{\sum K\left(\frac{x - x_i}{h}\right)(Y_{1i} + Y_{2i} + Y_{3i})}{\sum K\left(\frac{x - x_i}{h}\right)} = 1 \quad \longrightarrow \quad \text{Nadaraya–Watson estimator of the regression function (Kernel reg}$$

1.2 Nearest Neighbor Regression

$$\widehat{E(Y_j \mid x_1 \cdots x_d)} = \frac{\sum_{i:\, x_i \text{ one of the } k\text{-neighbors of } x} Y_{ji}}{k}, \qquad j = 1, 2, 3$$

$$\sum_j \widehat{E(Y_j \mid x)} = 1$$

1.3 Linear Regression

$$\widehat{E}(Y_1 \mid x_0) = x_0'(X'X)^{-1}X'Y_1, \qquad \widehat{E}(Y_2 \mid x_0) = x_0'(X'X)^{-1}X'Y_2, \qquad \widehat{E}(Y_3 \mid x_0) = x_0'(X'X)^{-1}X'Y_3$$

$$X = \begin{bmatrix} 1 & X_{11} & X_{21} & \cdots & X_{d1} \\ 1 & X_{12} & X_{22} & \cdots & X_{d2} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & X_{1n} & X_{2n} & \cdots & X_{dn} \end{bmatrix}, \qquad Y = \begin{bmatrix} Y_1 & Y_2 & Y_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 1 & 0 \end{bmatrix}$$

$$\hat{Y} = \begin{bmatrix} \hat{Y}_1 & \hat{Y}_2 & \hat{Y}_3 \end{bmatrix}, \qquad \hat{Y}_1 = X(X'X)^{-1}X'Y_1, \quad \hat{Y}_2 = X(X'X)^{-1}X'Y_2, \quad \hat{Y}_3 = X(X'X)^{-1}X'Y_3$$

We have $Y\mathbf{1} = \mathbf{1}$. We need to check $\hat{Y}\mathbf{1} = \mathbf{1}$?

$$\hat{Y} = (\hat{Y}_1, \hat{Y}_2, \hat{Y}_3) = X(X'X)^{-1}X'(Y_1, Y_2, Y_3) = X(X'X)^{-1}X'Y$$

$$\hat{Y}\mathbf{1} = X(X'X)^{-1}X'Y\mathbf{1} = X(X'X)^{-1}X'\mathbf{1} = \mathbf{1}$$

2 Single Layer Perceptron (SLP)

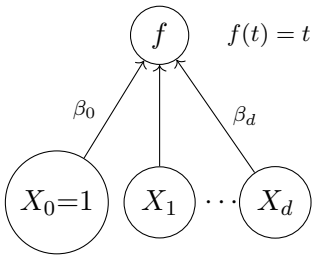


Figure 1: *
Linear model as a special case of SLP: $f(t) = t$

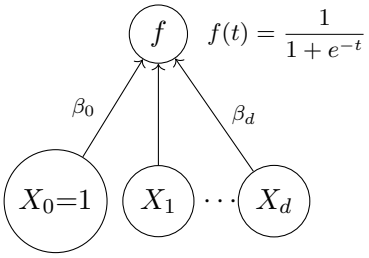


Figure 2: *
SLP with sigmoid activation, $t = \beta_0 + \sum_j \beta_j X_j$

Linear model is a special case of SLP. Normal least square method is used to get $\hat{\beta}_0, \ldots, \hat{\beta}_d$.

In SLP with sigmoid activation. Here, the least square method does *not* have a closed form solution. One needs to use gradient descent or Newton–Raphson method for minimization of the squared error loss

$$\sum \Big[Y_i - \sigma\big(\beta_0 + \sum \beta_j X_{ij}\big) \Big]^2$$

Instead of squared error loss one can use the "Softmax" loss function. In that case also, gradient descent or Newton–Raphson method needs to be used for minimization.

2.1 Softmax Loss Minimization

$$Y_1, \ldots, Y_n \qquad \hat{Y}_1, \ldots, \hat{Y}_n$$

$$l(Y_1, \ldots, Y_n, \hat{Y}_1, \ldots, \hat{Y}_n) = -\sum_{i=1}^{n} \Big[Y_i \log \hat{Y}_i + (1 - Y_i) \log(1 - \hat{Y}_i) \Big]$$

$$= -\sum \Big\{ Y_i \log \sigma\Big(\beta_0 + \sum_j \beta_j X_{ij}\Big) + (1 - Y_i) \log \Big(1 - \sigma\big(\beta_0 + \sum_j \beta_j X_{ij}\big)\Big) \Big\}$$

$$= -\sum_{i=1}^{n} \left[Y_i \log \left(\frac{e^{\beta_0 + \sum_j \beta_j X_{ij}}}{1 + e^{\beta_0 + \sum_j \beta_j X_{ij}}} \right) + (1 - Y_i) \log \left(\frac{1}{1 + e^{\beta_0 + \sum_j \beta_j X_{ij}}} \right) \right]$$

$=$ negative of the log-likelihood function used in logistic regression.

What kind of classifier do we get from the SLP model with sigmoid transformation? (sq. error loss, softmax loss)

We classify an observation $x = (x_1, \ldots, x_d)$ to class-1 if

$$\hat{P}(Y{=}1 \mid x) = \frac{e^{\hat{\beta}_0 + \sum_j \hat{\beta}_j x_j}}{1 + e^{\hat{\beta}_0 + \sum_j \hat{\beta}_j x_j}} > 0.5 \iff \hat{\beta}_0 + \sum \hat{\beta}_j x_j > C_0 \quad (C_0 = 0 \text{ here})$$

(monotonically increasing function) for some C_0 (*linear classifier*).

To address this issue, multi layer perceptrons were developed.

3 Multi-Layer Perceptron (MLP)

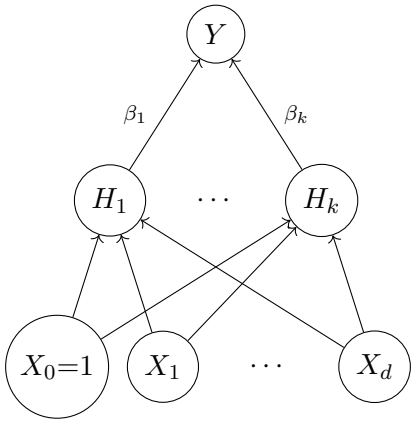


Figure 3: *

MLP model: input layer → hidden layer $(\sigma_1, \ldots, \sigma_k)$ → output (σ)

Input–output relationship:

$$H_1 = \sigma_1\Big(\alpha_{01} + \sum_j \alpha_{j1} X_j\Big), \quad H_2 = \sigma_2\Big(\alpha_{02} + \sum_j \alpha_{j2} X_j\Big), \quad \ldots, \quad H_k = \sigma_k\Big(\alpha_{0k} + \sum_j \alpha_{jk} X_j\Big)$$

$$Y = \sigma\Big(\beta_0 + \sum_{l=1}^{k} \beta_l H_l\Big)$$

$$\hat{Y}_r = \sigma\left[\beta_0 + \sum_{l=1}^{k} \beta_l\, \sigma_l\Big(\alpha_{0l} + \sum_{j=1}^{d} \alpha_{jl} X_{jr}\Big)\right], \qquad r = 1, 2, \ldots, n$$

$Y_1, \ldots, Y_n$: observed responses. $\hat{Y}_1, \ldots, \hat{Y}_n$: estimated responses.

We can use either the squared error loss or the softmax loss function and estimate the parameters by minimizing the chosen loss function.

$$\sum (Y_i - \hat{Y}_i)^2 \quad \text{— sq. error}$$

$$-\sum_{i=1}^{n} \Big[Y_i \log(\hat{Y}_i) + (1 - Y_i) \log(1 - \hat{Y}_i)\Big] \quad \text{— softmax loss}$$

$$\sum_{r=1}^{n} \left[Y_r - \sigma\left(\beta_0 + \sum_{l=1}^{k} \beta_l \sigma_l\Big(\alpha_{0l} + \sum_{j=1}^{d} \alpha_{jl} X_{jr}\Big)\right)\right]^2 \quad \text{squared error loss}$$

$$= \Psi(\beta_0, \beta_1, \ldots, \beta_k) = \Psi(\beta)$$

4 Gradient Descent Update

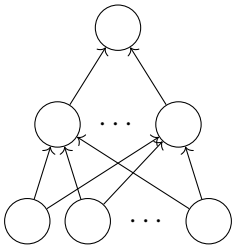


Figure 4: *

Network recap: parameters α_{jl} (input–hidden) and β_l (hidden–output)

Updated *old one*

$$\beta^{**} = \beta^{*} - \eta\, \nabla \Psi(\beta^{*})$$

Once these β_i's are estimated, we shall use the gradient descent method to modify the estimates of the α_{jl}'s.

5 Linear Model (Recap)

$$\underbrace{X_1, \ldots, X_d}_{\text{covariates}}, \qquad \underbrace{Y}_{\text{response variable}}$$

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_d X_{di} + \varepsilon_i, \qquad \varepsilon_i \text{ iid } N(0, \sigma^2), \quad (i = 1, \ldots, n)$$

- $Y \sim$ Normal distribution.
- $\mu = E(Y)$ is a linear function of the X_i's: $\mu = \beta_0 + \beta_1 X_1 + \cdots + \beta_d X_d$.
- $Var(Y) = \sigma^2$ has no relation (independent) with μ.

Lecture: Wednesday, October 14, 2026

6 Generalised Linear Model (GLM)

Linear model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_d X_{di} + \varepsilon_i, \qquad \varepsilon_i \text{ iid } N(0, \sigma^2)$$

- The response variable Y is normally distributed.
- $\mu = E(Y)$ is a linear function of $X_1, \ldots, X_d$.
- $\sigma^2 = Var(Y)$ is independent of μ.

But there are many cases where these conditions get violated. For example, let Y be the proportion of success in an experiment carried out at different sets of values for the covariates.

	X_1	X_2	$\cdots X_d$	No. of success Z	Prob. of success Y
n_1	X_{11}	X_{21}	$\cdots X_{d1}$	Z_1	
n_2					
$\vdots$					
n_k					
	Covariates			Response	

$$L = \prod_{i=1}^{K} \binom{n_i}{Z_i} p_i^{Z_i} (1 - p_i)^{n_i - Z_i}$$

Likelihood in terms of Y:

$$\prod_{i=1}^{K} \binom{n_i}{n_i Y_i} p_i^{n_i Y_i} (1 - p_i)^{n_i - n_i Y_i}$$

$$\mu \in (0, 1)$$

$$\eta = \text{lin. func. of the } X's = \beta_0 + \beta_1 X_1 + \cdots + \beta_d X_d \in (-\infty, \infty)$$

$$(-\infty, \infty) \longrightarrow (0, 1), \qquad \eta = \log \frac{p}{1 - p}$$

$$\mu = E(Y) = p \in (0, 1)$$

$$\mu(x_1, \ldots, x_d) = E(Y \mid x_1, \ldots, x_d) = p_i 1 \in (0, 1)$$

$$\eta = \beta_0 + \beta_1 x_1 + \cdots + \beta_d x_d \in (-\infty, \infty)$$

$\Phi^{-1}(p)$ — inverse of a cdf of a continuous r.v.

$$p = E(Y) = \mu, \qquad \eta = g(\mu), \qquad p = \mu = g^{-1}(\eta) = \frac{e^{\eta}}{1 + e^{\eta}}$$

If we take $\eta = g(\mu)$ where $g(t) = \log \dfrac{t}{1-t}$ (the logit function), then the likelihood can be written as

$$\prod_{i=1}^{K} \binom{n_i}{n_i Y_i} \left(\frac{e^{\eta_i}}{1+e^{\eta_i}}\right)^{n_i Y_i} \left(\frac{1}{1+e^{\eta_i}}\right)^{n_i - n_i Y_i}$$

$$= \prod_{i=1}^{n} \binom{n_i}{n_i Y_i} \left(\frac{e^{\beta_0+\beta_1 X_{1i}+\cdots+\beta_d X_{di}}}{1+e^{\beta_0+\beta_1 X_{1i}+\cdots+\beta_d X_{di}}}\right)^{n_i Y_i} \left(\frac{1}{1+e^{\beta_0+\beta_1 X_{1i}+\cdots+\beta_d X_{di}}}\right)^{n_i - n_i Y_i}$$

Note that if $n_i = 1$ for all i, then the likelihood is

$$= \prod_{i=1}^{K} \left(\frac{e^{\beta' x_i}}{1+e^{\beta' x_i}}\right)^{Y_i} \left(\frac{1}{1+e^{\beta' x_i}}\right)^{1-Y_i} \longrightarrow \text{the function that we get in the case of logistic regression.}$$

Here we can maximize the likelihood w.r.t. $\beta_0, \beta_1, \ldots, \beta_d$.
Let $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_d$ be the maximum likelihood estimates.

$$n_0,\ x_0 = (x_{01}, x_{02}, \ldots, x_{0d})$$

$$\hat{Y} = \frac{e^{\hat{\beta}' x_0}}{1+e^{\hat{\beta}' x_0}}, \qquad Z \sim Bin\left(n_0,\ \frac{e^{\hat{\beta}' x_0}}{1+e^{\hat{\beta}' x_0}}\right)$$

Example

Y = No. of goals scored in a soccer match.

$$\begin{aligned}
X_1 &= \text{eff. of player 1 in team 1}\\
X_2 &= \vdots\\
X_{11} &= \vdots \quad \text{(player 11, team 1)}\\
X_{12} &= \text{eff. of player 1 in team 2}\\
X_{13} &= \text{eff. of player 2 in team 2}\\
&\vdots\\
X_{22} &= \text{eff. of player 11 in team 2}
\end{aligned}$$

7 Poisson Regression

Match No.	$X_1, X_2, \ldots, X_{100}$	Y
1		
2		
$\vdots$		
K		

Assumption: $Y \sim Poisson(\lambda)$

$$\prod_{i=1}^{K} \frac{e^{-\lambda_i} \lambda_i^{Y_i}}{Y_i!}$$

Likelihood function:

$$\prod_{i=1}^{n} e^{-e^{\beta_0+\beta_1 X_{1i}+\cdots+\beta_d X_{di}}} \frac{\left(e^{\beta_0+\beta_1 X_{1i}+\cdots+\beta_d X_{di}}\right)^{Y_i}}{Y_i!}$$

$$\lambda = \mu = e^{\eta} = e^{\beta' x}$$

$$E(Y) = \mu = \lambda$$
$$\eta = \beta_0 + \beta_1 X_1 + \cdots + \beta_d X_d$$
$$\eta \in (-\infty, \infty), \qquad \mu \in (0, \infty)$$
$$\eta = \log \mu$$

Log-likelihood function:

$$\sum_{i=1}^{K} \Big[- e^{(\beta_0+\beta_1 X_{1i}+\cdots+\beta_d X_{di})} + Y_i(\beta_0 + \beta_1 X_{1i} + \cdots + \beta_d X_{di}) \Big] + \text{term indep. of } \beta$$

One needs to maximize it to get the estimates of the β's.
Given x_0: $\eta_0 = \beta_0 + \beta_1 x_{01} + \cdots + \beta_d x_{0d}$, $\lambda_0 = \mu_0 = e^{\eta_0}$, $Y \sim P(\lambda_0)$.

Y	$X_1, \ldots, X_d$
$\vdots$	$\vdots$

$$\prod_{i=1}^{K} e^{-\lambda} \frac{\lambda^{Y_i}}{Y_i!}$$

Example: (the response is 0–1 valued)

8 Logistic / Probit Regression (Bernoulli Case)

$X_1, \ldots, X_d$	Y
p_1	Y_1
p_2	Y_2
$\vdots$	$\vdots$
p_K	Y_K

Likelihood:

$$\prod_{i=1}^{K} p_i^{Y_i} (1 - p_i)^{1-Y_i}$$

$$\mu = E(Y) = p \in (0, 1), \qquad \eta = \beta_0 + \beta_1 X_1 + \cdots + \beta_d X_d \in (-\infty, \infty)$$

We need a function g such that $\eta = g(\mu)$:

$$g(t) = \log \frac{t}{1-t} \quad \text{logit function (logistic regression)}$$

$$* \quad \prod_{i=1}^{K} p_i^{Y_i}(1-p_i)^{1-Y_i} = \prod_{i=1}^{n} \Big(\Phi(\beta_0 + \beta_1 X_{1i} + \cdots + \beta_d X_{di})\Big)^{Y_i} \Big(1 - \Phi(\beta_0 + \beta_1 X_{1i} + \cdots + \beta_d X_{di})\Big)^{1-Y_i}$$

$$p = \mu = g^{-1}(\eta) = \Phi(\eta) = \Phi(x'\beta)$$

$$* \quad g(t) = \Phi^{-1}(t)$$

We can maximize it to get $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \ldots, \hat{\beta}_d$.
For a new observation $x_0 = (x_{01}, \ldots, x_{0d})$:

$$Y \sim Ber\big(\Phi(\hat{\eta}_0)\big) \quad \text{where} \quad \hat{\eta}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_{10} + \cdots + \hat{\beta}_d x_{0d}$$

8.1 Probit Regression

$X_1, \ldots, X_d$	Y
1	
2	
$\vdots$	
k	

$$Y \sim N(\mu, \sigma^2)$$

$$\prod_{i=1}^{K} \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{Y_i - \mu_i}{\sigma}\right)^2}\right] \qquad \text{assuming } \sigma_1 = \sigma_2 = \cdots = \sigma_K$$

$$\mu : E(Y) \in (-\infty, \infty)$$

9 Normal Case: Ordinary Least Squares as GLM

$$\eta = \beta_0 + \beta_1 X_1 + \cdots + \beta_d X_d, \qquad \eta = \mu, \qquad \eta = g(\mu) \text{ where } g(t) = t$$

$$\text{Likelihood} = \prod_{i=1}^{K} \left(\frac{1}{\sigma\sqrt{2\pi}}\right) e^{-\frac{1}{2}\left(\frac{Y_i - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_d X_{di}}{\sigma}\right)^2}$$

Maximization of this is equivalent to minimization of

$$\sum \big(Y_i - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_d X_{di}\big)^2$$

If σ_i's are not the same, then

$$\text{Likelihood} = \prod_{i=1}^{n} \left(\frac{1}{\sigma_i\sqrt{2\pi}}\right) e^{-\frac{1}{2}\left[\frac{Y_i - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_d X_{di}}{\sigma_i}\right]^2}$$

In that case, the maximization of the likelihood is equivalent to minimization of

$$\sum_{i=1}^{K} \left\{\frac{(Y_i - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_d X_{di})^2}{\sigma_i^2}\right\}$$

9.1 Summary of Link Functions

- Dist. for Y (Bernoulli, Binomial, Poisson or Normal).
- $\mu = E(Y), \quad \eta = \beta_0 + \beta_1 X_1 + \cdots + \beta_d X_d.$
- We need to look for a suitable link function g such that we can assume $\eta = g(\mu)$:

$$\begin{aligned} g(t) &= \log \frac{t}{1-t},\ \Phi^{-1}(t) \ \text{ for Bernoulli} && \longrightarrow \text{logistic regression / probit regression} \\ g(t) &= \log t \ \text{ for Poisson} && \longrightarrow \text{Poisson regression} \\ g(t) &= t \ \text{ for Normal} && \longrightarrow \text{usual least squares regression} \end{aligned}$$

Write down the likelihood in terms of η's and then maximize it w.r.t. β.

10 Exponential Family of Distributions

In GLM, we usually work with a special class of exponential distributions. We assume that the response variable Y has pdf/pmf of this form

$$f_\theta(y) = \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right\}$$

where ϕ is a scale parameter.

- If ϕ is known: $e^{y\theta/a(\phi)} \cdot e^{-b(\theta)/a(\phi)} \cdot e^{c(y,\phi)}$ — it belongs to the one-parameter exponential family.

$$A(\theta)B(x)e^{\tau(x)c(\theta)}$$

- If ϕ is not known, it may not belong to an exponential family of distributions.

Here we assume that ϕ and $a(\phi)$ are known to us. In most of the cases, $a(\phi)$ is of the form $a(\phi) = a_0\phi$.

10.1 Bernoulli

$$\begin{gathered} p^y(1-p)^{1-y} \\ \exp\big(y\log p + (1-y)\log(1-p)\big) \\ \exp\big[y\log p + \log(1-p) - y\log(1-p)\big] \\ = \exp\left[y\log\frac{p}{1-p} + \log(1-p)\right] \end{gathered}$$

10.2 Poisson

$$\begin{aligned} e^{-\lambda}\frac{\lambda^y}{y!} &= \exp\{-\lambda + y\log\lambda - \log y!\} \\ &= \exp\left\{\frac{y\log\lambda - \lambda}{1} - \log y!\right\} \end{aligned}$$

10.3 Normal

$$
\begin{aligned}
&\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(y-\mu)^2/\sigma^2} \\
&= \exp\left\{-\frac{1}{2\sigma^2}(y-\mu)^2 - \log(\sigma\sqrt{2\pi})\right\} \\
&= \exp\left\{-\frac{1}{2\sigma^2}\{y^2 - 2y\mu + \mu^2\} - \log(\sigma\sqrt{2\pi})\right\} \\
&= \exp\left[\frac{y\mu - \frac{1}{2}\mu^2}{2\sigma^2} - \log(\sigma\sqrt{2\pi}) - \frac{1}{2\sigma^2}y^2\right]
\end{aligned}
$$